



Bitchat and
chill...w/cashu

Welcome bitchatters

How do we move Bitcoin value when the internet is down?



- The Bitcoin protocol works anytime, anywhere but wallets need a network to broadcast transactions.



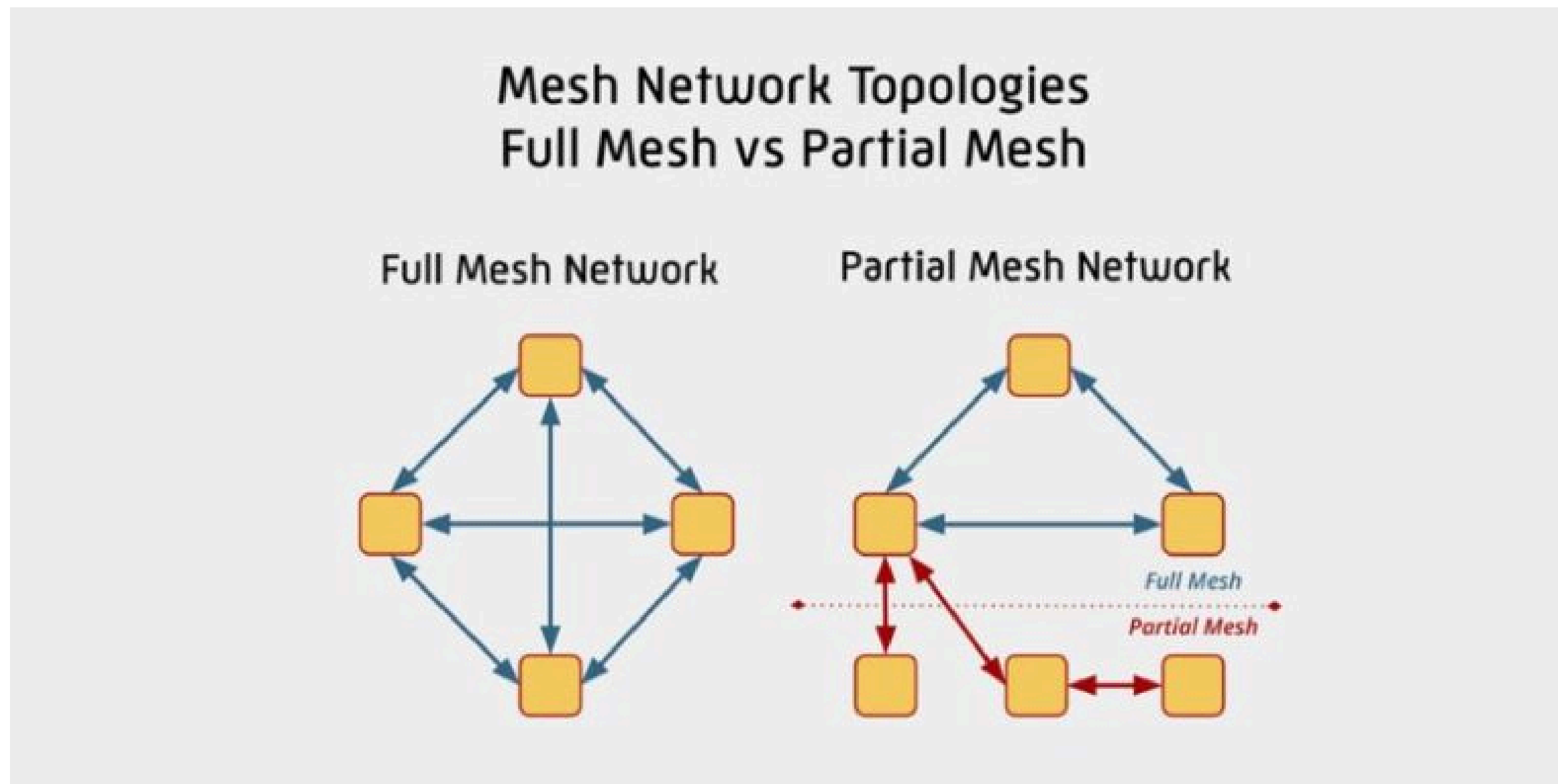
- Real-world “black-out” scenarios (natural disaster, censorship, festivals) still need peer-to-peer value transfer.



-What if we could combine offline mesh networking with privacy-preserving eCash so two phones can exchange value without ever touching the public internet?

What is a Mesh Network?

A mesh network is a distributed communications architecture in which every device (called a node) can both send and receive data and forward traffic for other nodes. Instead of relying on a single central hub (like a router or cell tower), the network builds many redundant paths, so messages can “hop” from node to node until they reach their destination.



Types of Mesh Networks

Category	Typical Radio / Transport	Example Protocols / Platforms	Typical Use-Case
Wi-Fi Mesh	2.4 / 5 GHz IEEE 802.11	802.11s, OpenMesh, Google Nest Wi-Fi, Ubiquiti AmpliFi	Home/office broadband extension, city-wide public Wi-Fi.
Bluetooth Mesh	Bluetooth Low Energy (BLE) 2.4 GHz	Bluetooth Mesh Profile, Briar (BLE mode), Bitchat BLE	Secure local chat, IoT command & control, short-range token exchange.
LoRa Mesh	LoRa (sub-GHz, 433/868/915 MHz)	Meshtastic, LoRa-Mesh, The Things Network (self-hosted)	Rural/remote sensor networks, long-range (km) low-bandwidth messaging.
Zigbee / Thread	2.4 GHz (IEEE 802.15.4)	Zigbee, Thread (Matter)	Smart-home lighting, low-power IoT devices.
Hybrid / Opportunistic	Multiple radios (BLE + Wi-Fi + Cellular + Ethernet)	Briar (BLE ↔ Wi-Fi-Direct ↔ Tor), CJDNS, Yggdrasil	Disaster-response communication, “offline-first” apps.
Satellite / High-Altitude	Radio links to balloons, cubesats	Project Loon, SpaceX Starlink mesh (future)	Global back-haul for remote mesh islands.

GoTenna



What: Commercial off-the-shelf “walkie-talkie” messenger.

Protocol: Proprietary 2.4 GHz FSK radio with a custom store-and-forward mesh; AES-256 encryption (handled internally, no user-set key).

How: Short packets are broadcast; an optional cloud can register devices, but the mesh works fully offline.

Pros: Plug-and-play UI; native iOS + Android apps; works without any network.

Cons: Closed firmware; real-world throughput $\approx 1-2$ KB/s; relatively high per-unit price. Original GoTenna version no longer supported. New versions marketed to LE, Govt, Military



Meshtastic

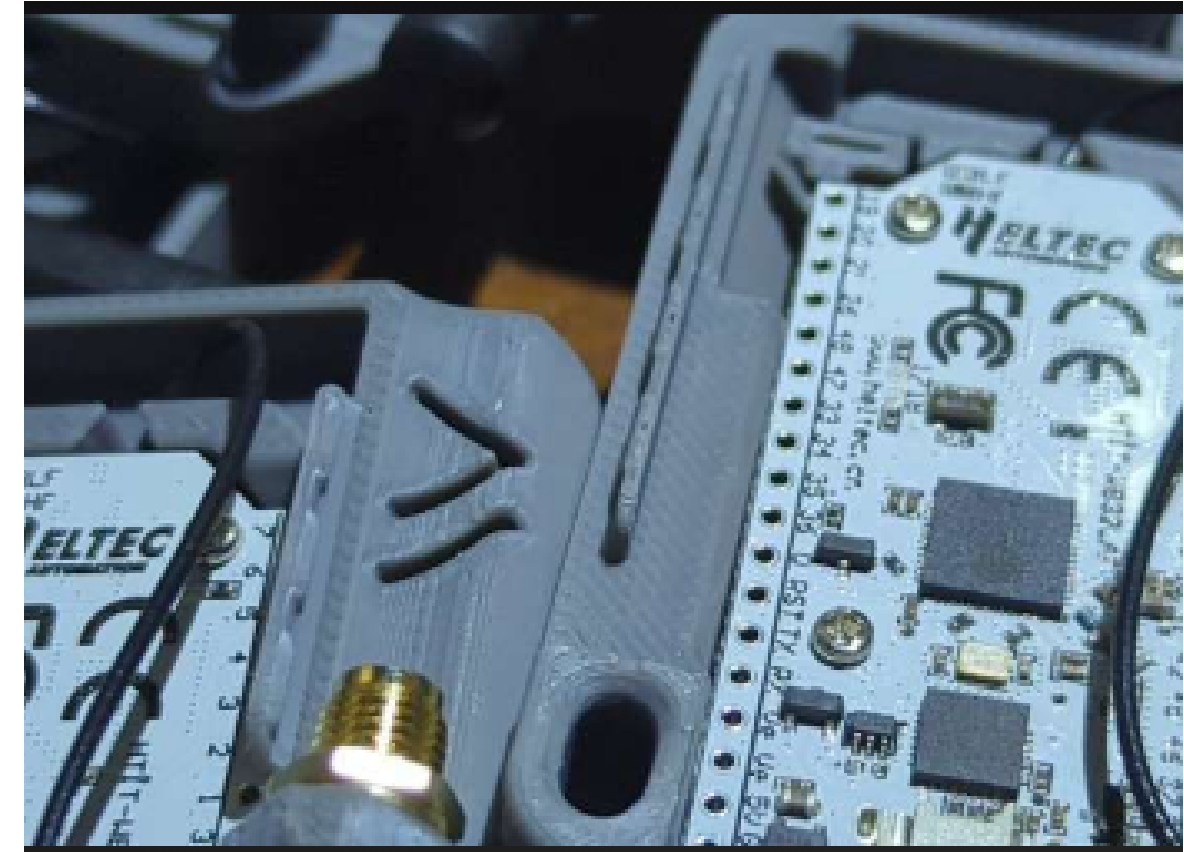
What: Open-source long-range radio (“LoRa”) firmware for cheap hobby boards (Heltec LoRa 32).

Protocol: LoRa PHY + “Meshtastic Mesh” (protobuf packets); AES-256 encryption with per-node session keys.

How: 1–5 km line-of-sight (up to 10 km with a high-gain antenna); any node can run an MQTT (Message Queuing Telemetry Transport)/Internet bridge.

Pros: DIY-friendly; low-cost hardware; long range; active community; on-device encryption by default.

Cons: Requires flashing firmware; max user payload $\approx$ 242 B; latency can be seconds-to-minutes because of flood-and-prune routing.



Briar

What: Decentralized Android messenger designed to work offline: it syncs messages directly between devices via Bluetooth and Wi-Fi (and via Tor when the internet is available).

Protocol: Briar's Bramble protocol is end-to-end encrypted, peer-to-peer sync over Bluetooth or Wi-Fi (incl. Wi-Fi Direct); Tor when online. Store-and-forward with rotating session keys for forward secrecy.

How: Add contacts in person (QR). When nearby, devices sync encrypted messages peer-to-peer over Bluetooth or Wi-Fi (incl. Wi-Fi Direct/local hotspot); messages spread as contacts meet (store-and-forward).

Pros: No central servers; offline sync over Bluetooth/Wi-Fi; open-source; forward secrecy via rotating keys.

Cons: Android-only, needs dense user base for multi-hop, BLE bandwidth ~100 KB/s



Bitchat

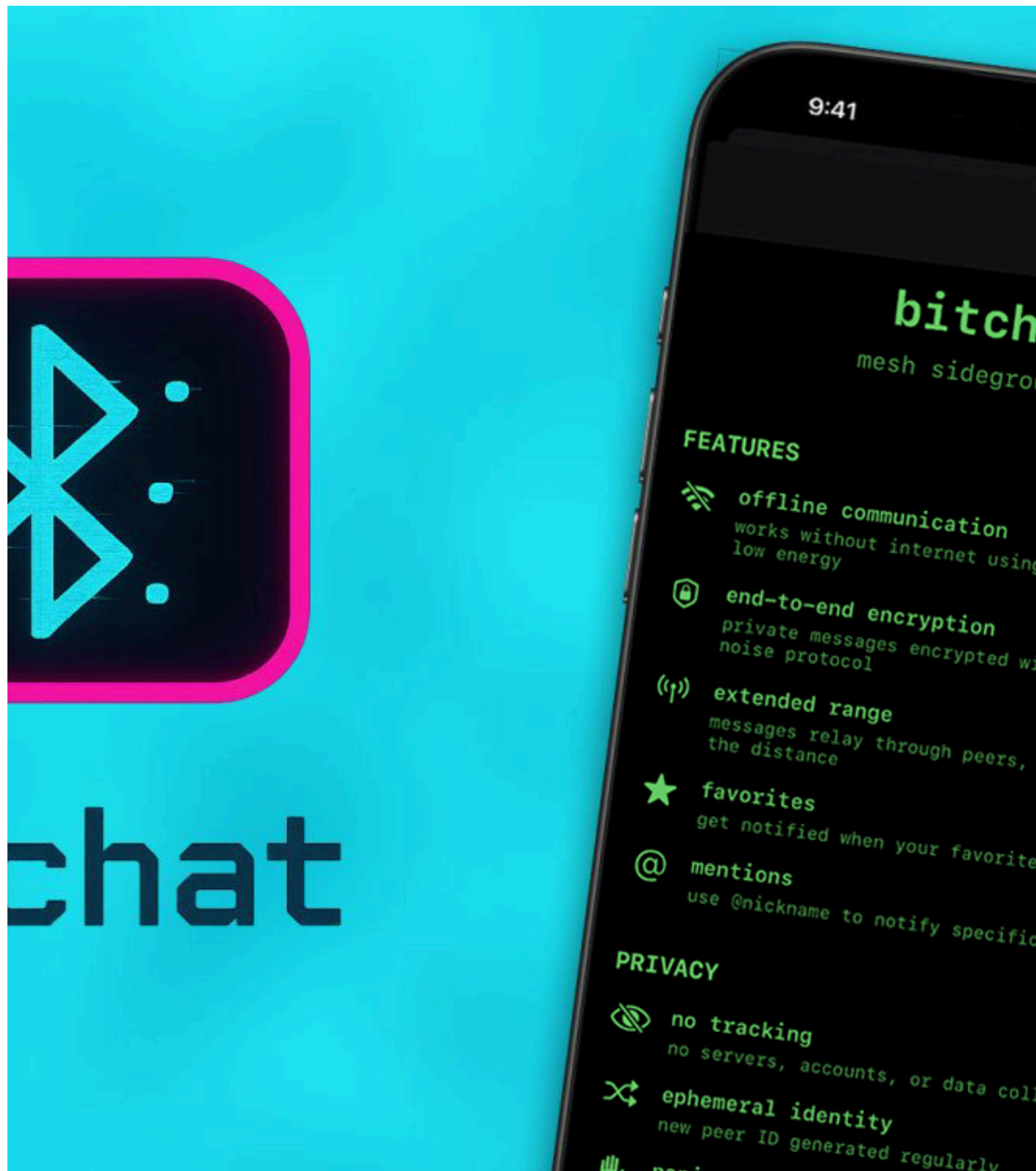
What: Jack Dorsey's encrypted messenger that can operate over BLE when no internet is present.

Protocol: BLE mesh with automatic peer discovery; devices act as both central and peripheral. Supports multi-hop relays (TTL), de-duplication, and automatic fragmentation of longer messages. Private DMs use the Noise Protocol for end-to-end encryption; no independent security audit yet.

How: Nearby devices discover and connect over BLE, exchange small encrypted payloads, and relay them hop-by-hop. Messages that exceed the BLE MTU are split and reassembled automatically.

Pros: Minimal setup, works offline, iOS + Android availability, E2E for private messages.

Cons: Very new – limited public audits, payload ceiling ~512 B, reliance on BLE stack quirks.





Why CashU + eCash fits perfectly on a mesh

- eCash = privacy-preserving bearer tokens (no account, no KYC).
- CashU issues those tokens from a mint; each token is a self-contained QR / base-64 string (~280-350 B).
- Workflow: Mint → Sender → Receiver (offline) → Mint (redeem). Only the mint needs internet.
- Why it matters:
 - a. Tiny payload → fits in BLE, LoRa, or Bluetooth messages.
 - b. No on-chain transaction until redemption (truly “offline”).
 - c. Same anonymity model as physical cash.

Demo: Cashu → Meshtastic (LoRa)

1. Mint a 21 sat token
2. Copy token text to clipboard
3. Phones linked to Meshtastic nodes (BLE); same channel
4. Send token text via Meshtastic (auto-split if long)
5. Redeem later when online

CashU ↔ Briar (BLE) – From Mint to Pocket

1. Mint a 21 sat token
2. Copy token text to clipboard
3. Open Briar on Phone A → enable Offline/BLE mode.
4. Scan or paste the token into a Briar chat and hit Send.
5. Phone B receives the encrypted message (BLE mesh hop if needed).
6. Redeem token with CashU wallet (CLI or mobile) → mint credits the balance.

Demo: CashU ↔ Bitchat (BLE) – From Mint to Pocket

1. Mint a 21 sat token
2. Copy token text to clipboard
3. Open Bitchat and enable bluetooth
4. Scan / paste the token into a Bitchat message and hit Send.
5. Phone B receives the encrypted message (BLE mesh hop if needed).
6. Redeem token with CashU wallet (CLI or mobile) → mint credits the balance.

Where Do We Go From Here?

Trust – “Who do we have to trust in this flow? The mint, the phones, the mesh?”

Scalability – “How many hops could a BLE mesh realistically support in a crowded event?”

Local Mint – “What would a community-run CashU mint look like? What are the risks?”

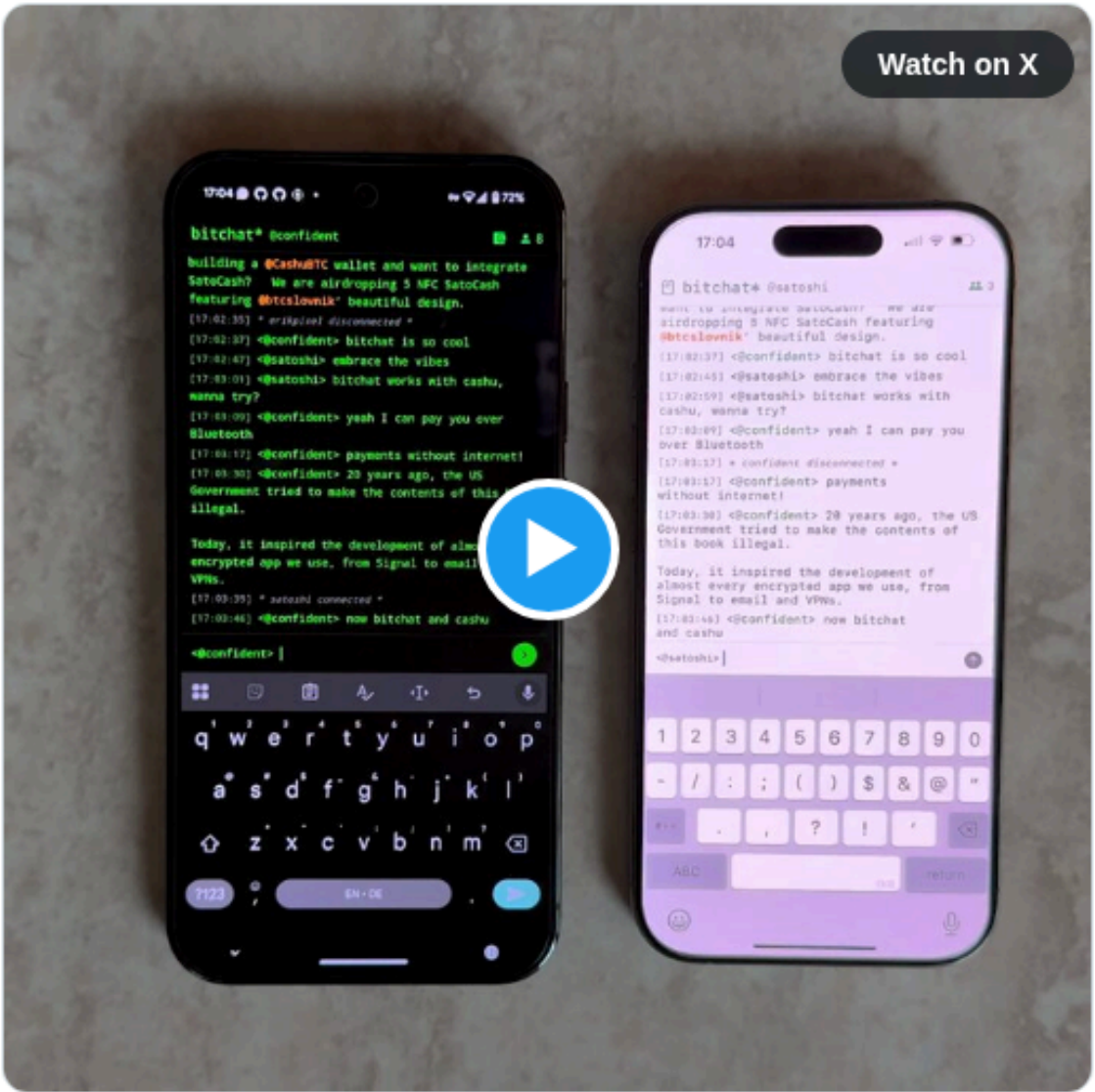
Alternative Meshes – “Could we combine BLE with LoRa (Meshtastic) for longer reach? What would change?”

User Experience – “What UI features would make a ‘mesh-first’ Bitcoin wallet feel native?”

sending bitcoin over bluetooth between bitchat android and iphone. both have a native cashu ecash wallet built in.

the ecash travels directly from phone to phone. the sender needs no internet. like instant and untraceable digital cash.

work in progress. it's going to be insane.



5:48 PM · Jul 17, 2025



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Quick demo of Samiz + Citrine + Jumble working with 3 phones sharing notes via BLE:

